DAY2 7/10/26

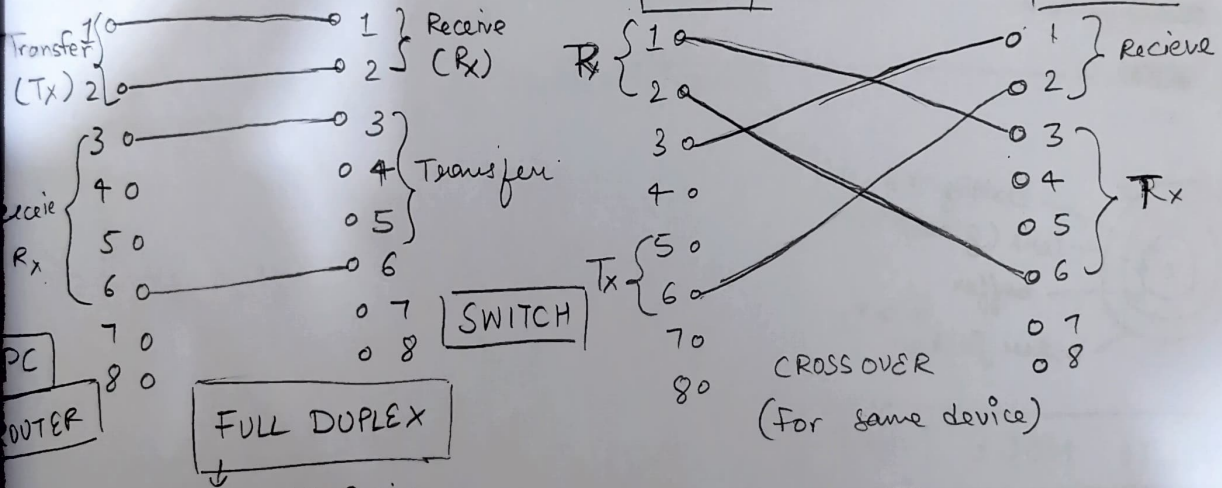
ETHERNET :- Collection of Network Protocols/standards
→ both are included
- physical standards
- Logical standards

- Data is transferred in bits in second
- Defined in IEEE 802.3 standards in 1983
- Copper Cable standards ;-

Speed	Common Name	IEEE Name	Informal Name	Max Length
10 Mbps	Ethernet	802.3i	10 Base T	100m
100 Mbps	Fast Ethernet	802.3u	100 Base T (2 pairs UTP)	
1 Gbps	Gigabit Ethernet	802.3ab	1000 Base T	
10 Gbps	10 Gig Ethernet	802.3an	10G Base T (4 pairs UTP)	

- UTP (unshielded Twisted Pairs) ⇒ 4 pairs ⇒ 8 wires
  - unshielded: no metal protection
  - Twisted: protects from EMI

for 10 & 100 Base T



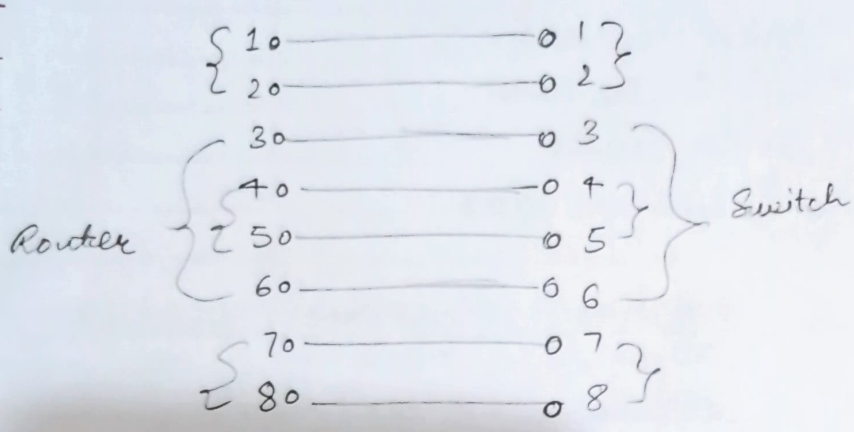
FULL DUPLEX
↓
Can send & Recieve data at same type

DEVICE TYPE	Tx	Rx
Router	1 & 2	3 & 6
Fire wall	1 & 2	3 & 6
PC	1 & 2	3 & 6
Switch	3 & 6	1 & 2

↓
10 & 100 Base-T

AUTO MDI-X
↓
Cables are adjusted automatically !!

UTP (1000BASET & 10GBASET)

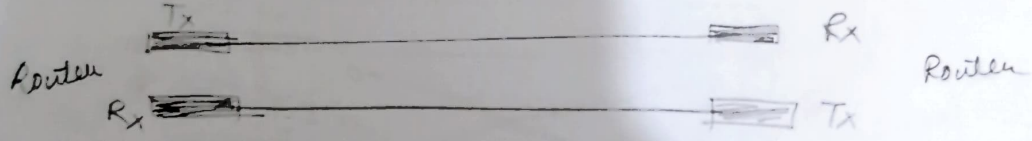


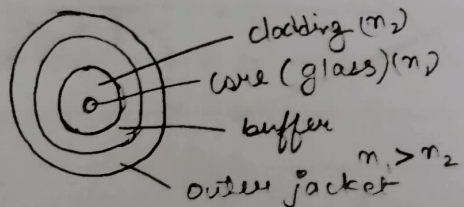
Each is Bidirectional

OPTICAL FIBRE:-

SFP Transceiver in RJ45

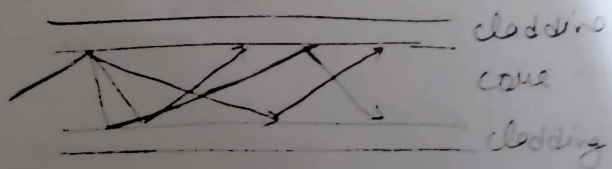
Small form factor pluggable





MULTI MODE FIBRE

- Core Φ is bigger
- Allows multiple angles TIR
- Longer cables as compared to UTP
- Cheap



SINGLE MODE FIBRE



- Core Φ is thin
- Light enters at single angle (mode) from LASER based transmitters
- Allows longest cable length.
- expensive

INFORMAL NAME	IEEE name	Speed	Cable Type	Max Length
1000 Base-LX	802.3z	1Gbps	MM or SM	550m (MM) 5km (SM)
10GBase SR	802.3ae	10Gbps	MM	400m
10GBase LR	802.3ae		SM	10km
10GBase ER	802.3ae		SM	10km